

wine_analysis

May 3, 2025

0.0.1 Setup

- Import necessary libraries.
- Load the red and white wine datasets from CSV files.
- Set up Seaborns style for clean, consistant visuals throughout.

```
[1]: # Imports
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Load data
wine_red = pd.read_csv("../datasets/winequality-red.csv", sep=";")
wine_white = pd.read_csv("../datasets/winequality-white.csv", sep=";")
wine_red.columns = wine_red.columns.str.strip()
wine_white.columns = wine_white.columns.str.strip()

# Set global style
sns.set_theme(style="whitegrid", font_scale=1.2)
palette_red = sns.color_palette("Reds_r", as_cmap=True)
palette_white = sns.color_palette("Blues", as_cmap=True)
```

0.0.2 Task 1A: Wine Quality Distributions

This side-by-side bar chart shows the distribution of quality scores for both wine types.

Note: The white wine dataset is significantly larger, which contributes to the higher overall bar heights.

```
[2]: # For combined plot: red vs white
fig, axes = plt.subplots(1, 2, figsize=(12, 5))

# Plots

# Red plot
sns.countplot(data=wine_red, x="quality",
              hue="quality", ax=axes[0], palette="Reds_r", legend=False)
axes[0].set_title("Red Wine Quality", fontsize=14)
```

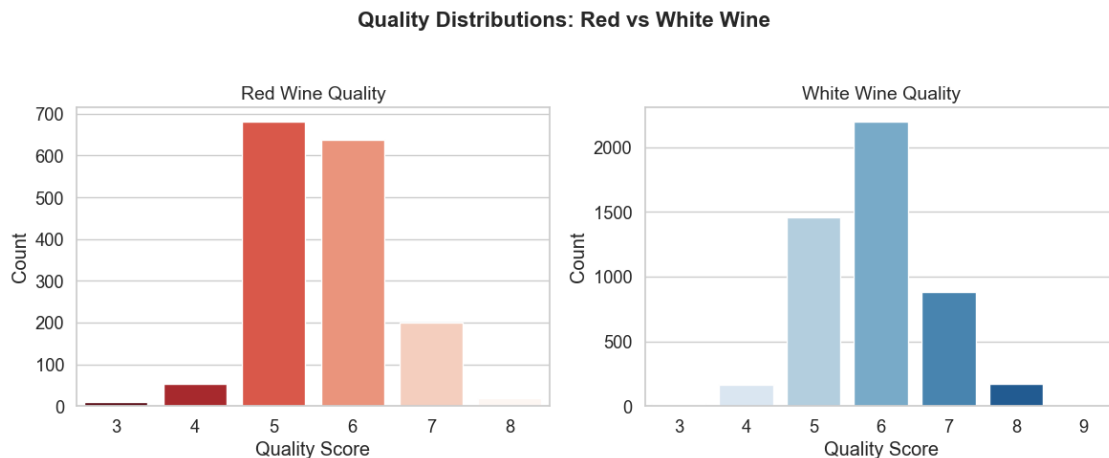
```

axes[0].set_xlabel("Quality Score")
axes[0].set_ylabel("Count")

# White plot
sns.countplot(data=wine_white,
              ↪x="quality", hue="quality", ax=axes[1], palette="Blues", legend=False)
axes[1].set_title("White Wine Quality", fontsize=14)
axes[1].set_xlabel("Quality Score")
axes[1].set_ylabel("Count")

# Both plot
plt.suptitle("Quality Distributions: Red vs White Wine", fontsize=16,
              ↪weight="bold")
plt.tight_layout(rect=[0, 0, 1, 0.95])
plt.savefig("../figures/wine_quality_comparison.png", dpi=300)
plt.show()

```



0.0.3 Red Wine Quality Distribution

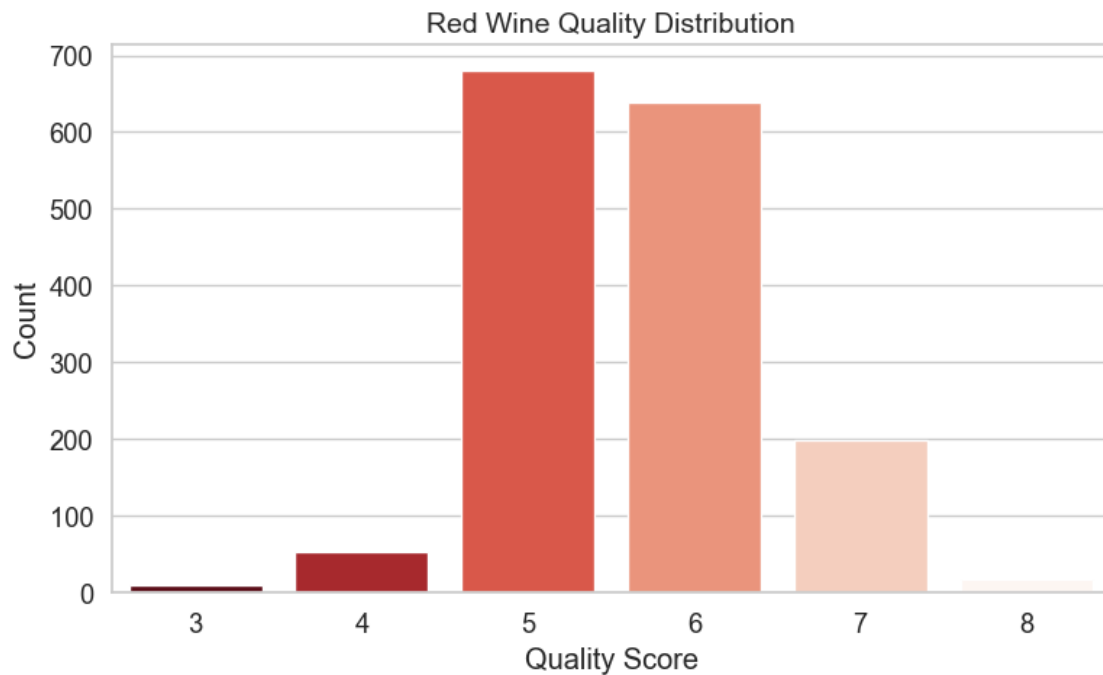
This plot shows only the red wine data. The distribution is narrow, mostly between scores 5 and 6.

```

[3]: # Red wine plot
plt.figure(figsize=(8, 5))
sns.countplot(data=wine_red, x="quality", hue="quality", palette="Reds_r",
              ↪legend=False)
plt.title("Red Wine Quality Distribution", fontsize=14)
plt.xlabel("Quality Score")
plt.ylabel("Count")
plt.tight_layout()
plt.savefig("../figures/red_wine_quality.png", dpi=300)

```

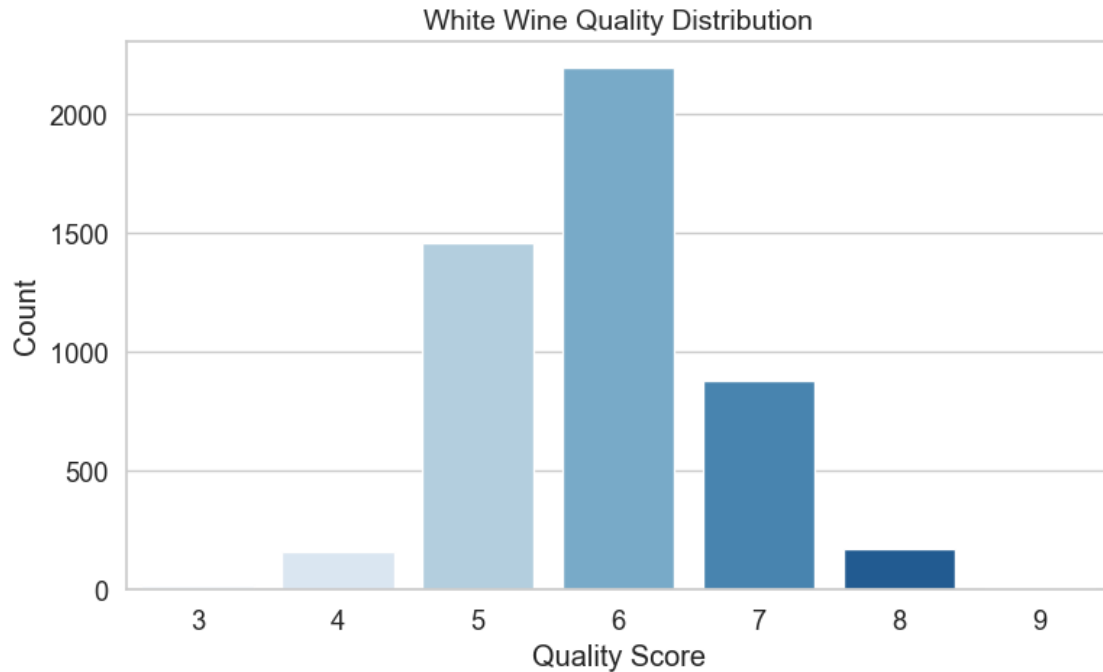
```
plt.show()
```



0.0.4 White Wine Quality Distribution

The white wine data spans a wider quality range and has more high-quality points (e.g. 7, 8, 9).

```
[4]: # White wine plot
plt.figure(figsize=(8, 5))
sns.countplot(data=wine_white, x="quality", hue="quality", palette="Blues",
              legend=False)
plt.title("White Wine Quality Distribution", fontsize=14)
plt.xlabel("Quality Score")
plt.ylabel("Count")
plt.tight_layout()
plt.savefig("../figures/white_wine_quality.png", dpi=300)
plt.show()
```



0.1 End: Task 1A

0.1.1 Summary

In this section of the project, I carried out the exploratory data analysis (EDA) necessary to begin understanding what affects perceived wine quality. As per the project specification:

I imported and explored both datasets (winequality-red.csv and winequality-white.csv), which contain attributes and quality ratings for red and white wines, and generated plots to visualise how quality scores are distributed across both wine types.

These visualisations show: - Red wine quality is tightly clustered around scores 5 and 6. - White wine quality is more spread out, with more high-rated samples (7 and 8), and a longer right tail. - There are over 3x as many white wine samples (4899 vs. 1599).

0.2 Begin Task 1B

0.2.1 Categorise Alcohol Levels

`add_alcohol_category` is a helper function to assign each sample an alcohol category: low, mid, or high, based on its deviation from the mean alcohol content.

```
[5]: # Add alcohol category
def add_alcohol_category(df):
    mean = df["alcohol"].mean()
    std = df["alcohol"].std()
```

```
def categorize(alc):
    if alc < mean - std:
        return "low"
    elif alc > mean + std:
        return "high"
    else:
        return "mid"

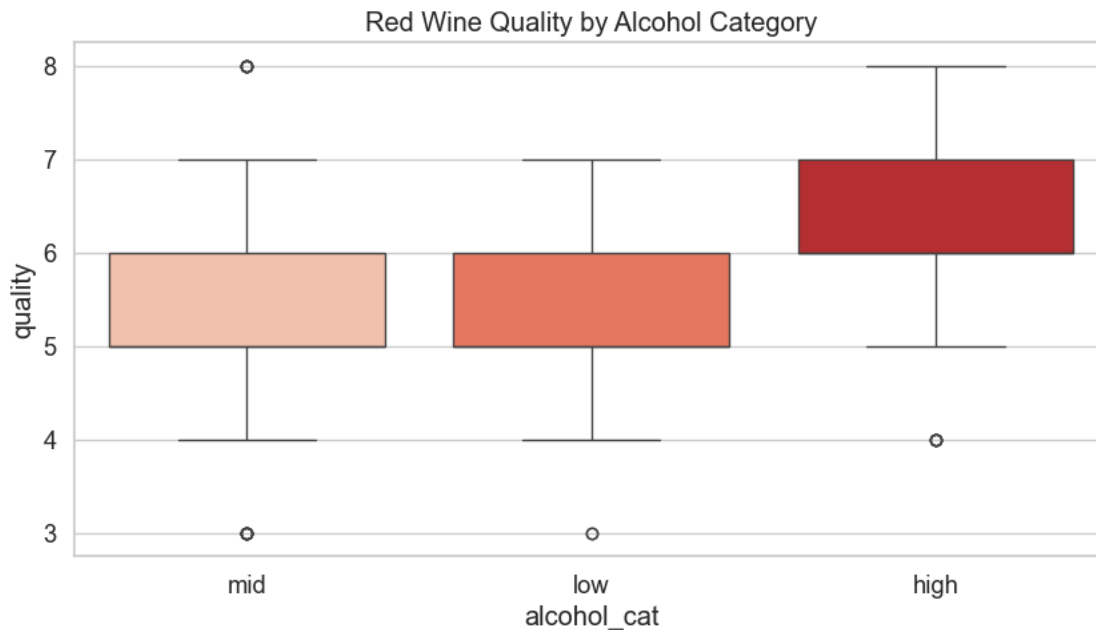
return df.assign(alc_cat=df["alc"].apply(categorize))
```

0.3 Red Wine: Quality by Alcohol Category

This plots how red wine quality is distributed across alcohol levels. The boxplot shows clear medians and the spread of quality values within each alcohol group.

```
[6]: # Apply to both datasets
wine_red = add_alcohol_category(wine_red)
wine_white = add_alcohol_category(wine_white)

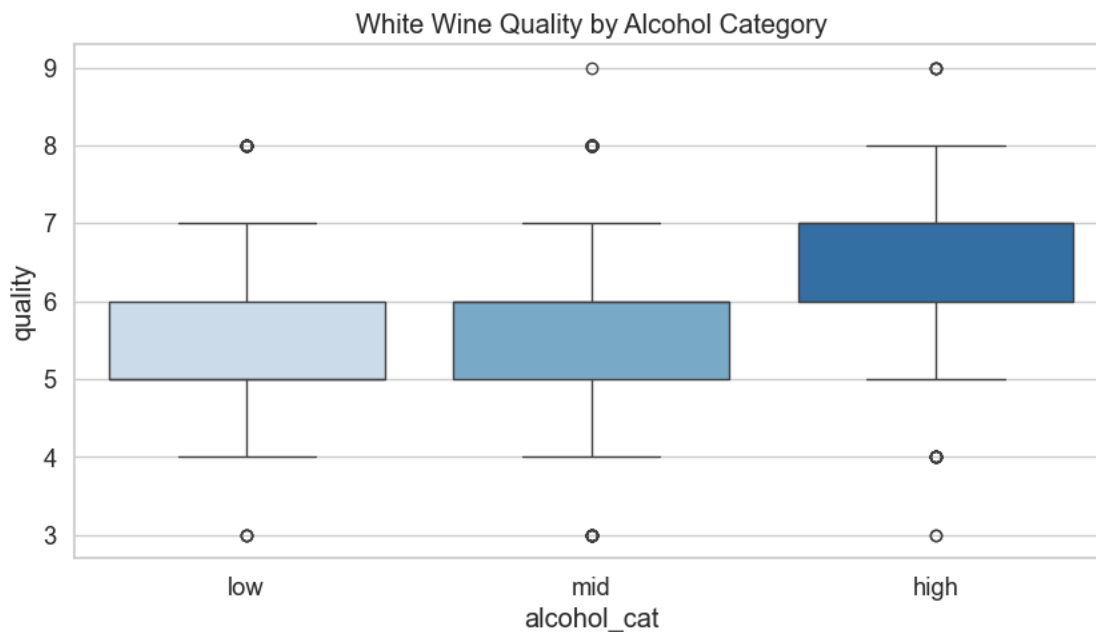
# Red wine
plt.figure(figsize=(10, 5))
sns.boxplot(data=wine_red, x="alc_cat", y="quality", palette="Reds",
            hue="alc_cat")
plt.title("Red Wine Quality by Alcohol Category")
plt.savefig("../figures/red_quality_by_alcohol.png")
plt.show()
```



0.4 White Wine: Quality by Alcohol Category

Repeat the same visual analysis for the white wine dataset.

```
[7]: # White wine
plt.figure(figsize=(10, 5))
sns.boxplot(data=wine_white, x="alcohol_cat", y="quality", palette="Blues",
            hue="alcohol_cat")
plt.title("White Wine Quality by Alcohol Category")
plt.savefig("../figures/white_quality_by_alcohol.png")
plt.show()
```



0.5 End: Task 1B

0.5.1 Summary

In this task, I generated box plot models visualising the wines perceived quality in regards to the alcohol content to identify if there was any correlation or relationship between the pair.

These visualisations show:

- Wines with higher alcohol content tend to receive higher quality scores for both red and white varieties.
- This supports a positive relationship between alcohol level and perceived wine quality.
- This feature will likely be useful for modelling in Task 3.